

Radial orthogonality does not imply dual 1-conformal flatness

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Abstract

We give a negative answer to Kurose’s Question 3(e) in the 1998 collection *Open Problems in Affine Differential Geometry and Related Topics*. For the product $S^2(1) \times \mathbb{R}$ with its product metric and Levi–Civita connection, every radial orthogonal distribution in that question is integrable, but the dual statistical structure is nowhere locally 1-conformally flat. Integrability follows from the first variation of energy; nonflatness follows from two curvature evaluations. A conformal metric and projective connection change also give an example with nowhere zero cubic tensor. The same obstruction works in every dimension at least three. The proof is independent of computation.

1 The question and the counterexample

A statistical manifold (M, ∇, h) consists here of a positive-definite metric h and a torsion-free connection ∇ such that ∇h is totally symmetric. The dual connection ∇^* is determined by

$$Xh(Y, Z) = h(\nabla_X Y, Z) + h(Y, \nabla_X^* Z).$$

Fix $p \in M$ and a ∇ -convex neighborhood U of p . For $q \in U \setminus \{p\}$, let $\gamma_q : [0, 1] \rightarrow U$ be the affinely parametrized ∇ -geodesic from p to q . Set

$$D_q^p = \{w \in T_q M : h_q(\dot{\gamma}_q(1), w) = 0\}. \quad (1)$$

Only the velocity line matters, so a regular reparametrization does not change this distribution.

Question 3(e) of [1, p. 126] asks whether integrability of every distribution (1) implies that (M, ∇^*, h) is 1-conformally flat. This is the printed question indexed as AMR-059-0011 (6000011). It excludes the center p and does not exclude Levi–Civita connections; item 3(a) explicitly treats those as statistical structures.

Following [2, p. 428], a 1-conformal change is

$$\hat{h} = e^\varphi h, \quad \hat{\nabla}_X Y = \nabla_X Y - h(X, Y) \operatorname{grad}_h \varphi. \quad (2)$$

The structure is 1-conformally flat if such a change makes $\hat{\nabla}$ flat locally at every point. This is a condition on the statistical connection, distinct from ordinary conformal flatness of h . We use $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$.

Theorem 1. *Let $M = S^2(1) \times \mathbb{R}$, $h = g_{S^2(1)} + dt^2$, and $\nabla = \nabla^h$. Every distribution (1) is integrable, for every center and every convex neighborhood on which it is defined. Nevertheless (M, ∇^*, h) is not 1-conformally flat on any nonempty open subset. The same conclusions hold for $S^2(1) \times \mathbb{R}^{n-2}$ in every dimension $n \geq 3$.*

2 An intrinsic proof

Lemma 1. *For any Riemannian manifold with its Levi-Civita connection, every distribution (1) is integrable.*

Proof. On a convex neighborhood define the smooth endpoint energy

$$E_p(q) = \frac{1}{2} \int_0^1 h(\dot{\gamma}_q(u), \dot{\gamma}_q(u)) du.$$

For an endpoint variation $q(s)$ with $q'(0) = w$, put $F(s, u) = \gamma_{q(s)}(u)$, $T = \partial_u F$, and $V = \partial_s F$. Metric compatibility, torsion-freeness, and the geodesic equation yield

$$\begin{aligned} dE_p|_q(w) &= \int_0^1 h(\nabla_V T, T) du = \int_0^1 h(\nabla_T V, T) du \\ &= [h(V, T)]_0^1 - \int_0^1 h(V, \nabla_T T) du = h_q(w, \dot{\gamma}_q(1)). \end{aligned}$$

The initial endpoint is fixed, so $V(s, 0) = 0$. For $q \neq p$ the terminal velocity is nonzero, and therefore $dE_p(\dot{\gamma}_q(1)) = h(\dot{\gamma}_q(1), \dot{\gamma}_q(1)) > 0$. Thus $D^p = \ker dE_p$ on $U \setminus \{p\}$, and its integral hypersurfaces are the regular level sets of E_p . On sufficiently small normal neighborhoods these are the geodesic spheres, since $E_p(q) = \frac{1}{2}d_h(p, q)^2$. The variation argument itself applies on any convex neighborhood in the question. $\square$

Lemma 2. *For a Levi-Civita statistical structure, local 1-conformal flatness requires an endomorphism A at each point satisfying*

$$R(X, Y)Z = h(Y, Z)A(X) - h(X, Z)A(Y). \quad (3)$$

Proof. Write $V = \text{grad}_h \varphi$ and $B(X, Y) = -h(X, Y)V$ for the difference tensor in (2). The curvature transformation law gives

$$\begin{aligned} \widehat{R}(X, Y)Z &= R(X, Y)Z + (\nabla_X B)(Y, Z) - (\nabla_Y B)(X, Z) \\ &\quad + B(X, B(Y, Z)) - B(Y, B(X, Z)) \\ &= R(X, Y)Z - h(Y, Z)A(X) + h(X, Z)A(Y), \end{aligned}$$

where $A(X) = \nabla_X V - h(X, V)V$. Indeed, $(\nabla_X B)(Y, Z) = -h(Y, Z)\nabla_X V$ and $B(X, B(Y, Z)) = h(Y, Z)h(X, V)V$. Setting $\widehat{R} = 0$ proves (3). $\square$

Proof of Theorem 1. Metric compatibility gives $\nabla h = 0$, so the structure is statistical and $\nabla^* = \nabla$. Lemma 1 gives the required integrability. At any point choose orthonormal vectors e_1, e_2 tangent to the unit sphere and e_3 tangent to the line. Product curvature gives

$$R(e_1, e_2)e_2 = e_1, \quad R(e_1, e_3)e_3 = 0.$$

If (3) held, these two equations would give respectively $A(e_1) = e_1$ and $A(e_1) = 0$, a contradiction. This excludes even an arbitrary endomorphism A , before imposing its differential form in Lemma 2. The contradiction holds at every point. Additional Euclidean factors leave these evaluations unchanged. $\square$

For comparison, Kurose's criterion [2, Proposition 1] identifies 1-conformal flatness with projective flatness of the dual connection with symmetric Ricci tensor. Here $\text{Ric} = \text{diag}(1, 1, 0)$ in the chosen frame. The projective Weyl tensor is

$$W(X, Y)Z = R(X, Y)Z - \frac{1}{2}\{\text{Ric}(Y, Z)X - \text{Ric}(X, Z)Y\}, \quad W(e_3, e_1)e_1 = -\frac{1}{2}e_3 \neq 0.$$

This is an independent obstruction. Neither this criterion nor a classification theorem is needed for the proof above.

3 A non-self-dual example

The failure is not confined to zero cubic tensor. Let $h_0 = g_{S^2(1)} + dt^2$, $\nabla^0 = \nabla^{h_0}$, and define

$$h_1 = e^t h_0, \quad \nabla_X^1 Y = \nabla_X^0 Y + dt(X)Y + dt(Y)X. \quad (4)$$

This is a standard projective change of connection, combined with a conformal change of metric. Such mechanisms belong to the theory of divisible cubic forms; see, for example, [3, §3.2]. We use them only to remove the possible restriction to self-dual examples.

Proposition 1. *The structure (M, ∇^1, h_1) has nowhere zero cubic tensor and integrable radial distributions for every center. Its dual is nowhere locally 1-conformally flat.*

Proof. The connection is torsion-free, and direct differentiation gives

$$(\nabla_X^1 h_1)(Y, Z) = -dt(X)h_1(Y, Z) - dt(Y)h_1(X, Z) - dt(Z)h_1(X, Y). \quad (5)$$

This is symmetric and has value $-3e^t$ on $(\partial_t, \partial_t, \partial_t)$. The connections ∇^1 and ∇^0 have the same unparametrized geodesics:

$$\nabla_{\dot{\gamma}}^1 \dot{\gamma} = \nabla_{\dot{\gamma}}^0 \dot{\gamma} + 2dt(\dot{\gamma})\dot{\gamma}.$$

The additional acceleration is tangent to the curve and is removed by a regular reparametrization. More explicitly, if u is a ∇^0 -affine parameter then a ∇^1 -affine parameter s satisfies $ds/du = Ce^{2t(\gamma(u))}$ with $C > 0$. This works on each compact geodesic segment. It identifies the unique connecting paths in the corresponding convex neighborhoods. Since h_1 and h_0 have the same orthogonality relation, their radial distributions agree under this identification, so Lemma 1 applies.

The defining duality identity gives

$$(\nabla^1)_X^* Y = \nabla_X^0 Y - h_0(X, Y)\partial_t.$$

For any local function ψ , its 1-conformal change is

$$\begin{aligned} \widehat{\nabla}_X Y &= (\nabla^1)_X^* Y - h_1(X, Y) \text{grad}_{h_1} \psi \\ &= \nabla_X^0 Y - h_0(X, Y) \text{grad}_{h_0}(t + \psi). \end{aligned}$$

If this were flat, it would be a flat 1-conformal change of (∇^0, h_0) , contrary to Theorem 1. □

Verification, attribution, and scope

The obstruction is elementary and the ingredients are classical. The contribution asserted here is the explicit negative answer to the printed Question 3(e), together with a short direct verification. A focused literature audit found no earlier explicit resolution, but does not establish historical priority. In particular, the live problem database’s annotations could not be checked; its identifier is only a cross-reference. We do not claim that the Gauss lemma, the curvature criterion, or the construction in (4) is new.

The accompanying standard-library Python checker solves the complete pointwise linear system (3) in exact rational arithmetic. A separately implemented SymPy checker derives the Christoffel symbols, curvature, transformation law, and cubic and dual tensors from $h_0 = d\theta^2 + \sin^2 \theta d\phi^2 + dt^2$. These checks supplement the proof; they do not replace its quantified first-variation argument or constitute a proof-assistant formalization. No dimension-two claim or global foliation across cut loci is needed.

AI disclosure. The candidate argument was supplied by the author as AI-generated work. This manuscript, supporting computations, source audit, and separate adversarial AI reviews were prepared with OpenAI Codex. This is an unrefereed preprint; AI reviews are not external human peer review.

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